ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION

A-LEVEL · PURE MATHEMATICS

MATHEMATICS 6042/1

PAPER 1 November 2022

3 hours

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

Additional materials: A calculator may be used unless a question forbids it.
INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

1. Find the first four terms in the expansion of $(1 + 2x)^6$ in ascending powers of x. **[8]**

2. The polynomial $p(x) = x^3 + ax^2 - 1x + 3$ has remainder 3 when divided by $(x - 1)$. Show this and find $p(-1)$. **[8]**

3. Differentiate $y = 6x^3 - 1x^2 + 1x - 1$. Hence find the gradient at $x = 1$. **[8]**

4. Find $\int (6x^2 - 1x + 1)\,dx$ and the definite integral from 0 to 1. **[8]**

5. Prove that $1 + \tan^2\theta = \sec^2\theta$. Hence solve $1 + \tan^2\theta = 4$ for $0° \le \theta \le 180°$. **[8]**

6. A is (1, 2), B is (7, 9).<<
>>(a) Midpoint [2]<<
>>(b) Distance AB [3]<<
>>(c) Equation of AB [3] **[8]**

7. An AP has first term 2 and common difference 2. Find (a) the 12th term [3] (b) S_{12} [5]. **[8]**

8. Solve $3^{2x} = 27$. Also simplify $\log_3 81$. **[8]**

9. $f(x) = 2x - 1$, $g(x) = x^2$.<<
>>(a) $fg(x)$ [2]<<
>>(b) $gf(x)$ [3]<<
>>(c) $f^{-1}(x)$ [3] **[8]**

10. Express $(5x+1)/((x+1)(x-2))$ in partial fractions. (Numbers may be checked by covering.) **[8]**

11. $A = (1, 1)$, $B = (4, 5)$. Find $|AB|$ and a unit vector in the direction of AB. **[8]**

12. A circle has radius 6 cm. A sector has angle 1 radians.<<
>>(a) Arc length [3]<<
>>(b) Sector area [5] **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).